

HAT-P-54b

R-1529

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_191127 · created 2026-08-01T17:57:37+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458814.8072 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.015
Transit depth (Rp/Rs)^2	2.62 +/- 0.5 [%]
Orbital Inclination (inc)	87.78 +/- 1.28
Airmass coefficient 1 (a1)	0.942 +/- 0.017
Airmass coefficient 2 (a2)	0.047 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.79 %
Best Comparison Star	#2 - [281, 148]
Optimal Method	PSF photometry
Transit Duration (day)	0.0699 +/- 0.0039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.162 ± 0.015 vs 0.1572 (deviation 0.32σ)
Duration fit vs published	0.0699 d vs 0.075 d (deviation 1.31σ)
Mid-time O-C	-0.6 min
Depth SNR	10.8
Residual scatter	1.79 %

Light curve

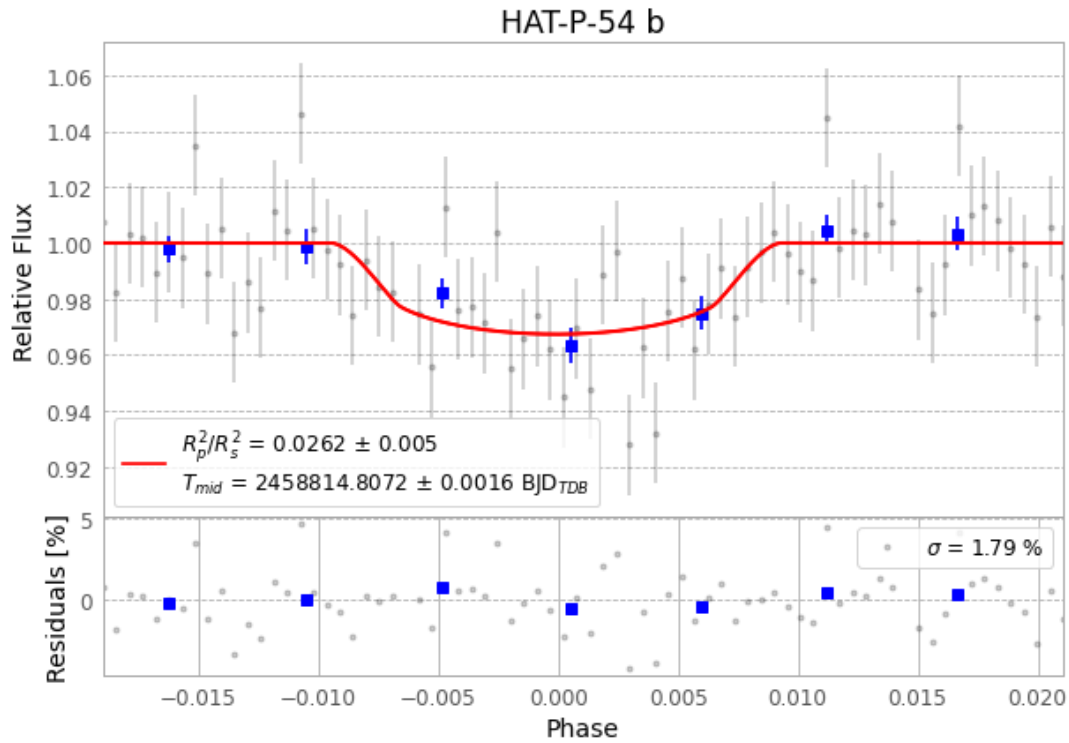


Figure 1 — FinalLightCurve_HAT-P-54 b_2019-11-26.png (SHA-256 1514b851edeb8ab8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [292, 223], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 92652cec5dbd81c1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), HATP-54191127053012.FITS → HATP-54191127090912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-191127.log (b5fcddb1e042...), FinalLightCurve_HAT-P-54 b_2019-11-26.png (1514b851edeb...), FinalLightCurve_HAT-P-54 b_2019-11-26.csv (d077f0f62d00...)

Compute resources

Wall time 205.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)